SaaS Customer Churn Analysis & Prediction Pipeline

An End-to-End Data Analytics & Machine Learning project demonstrating professional-level Data Engineering, Predictive Modeling, and Business Intelligence.

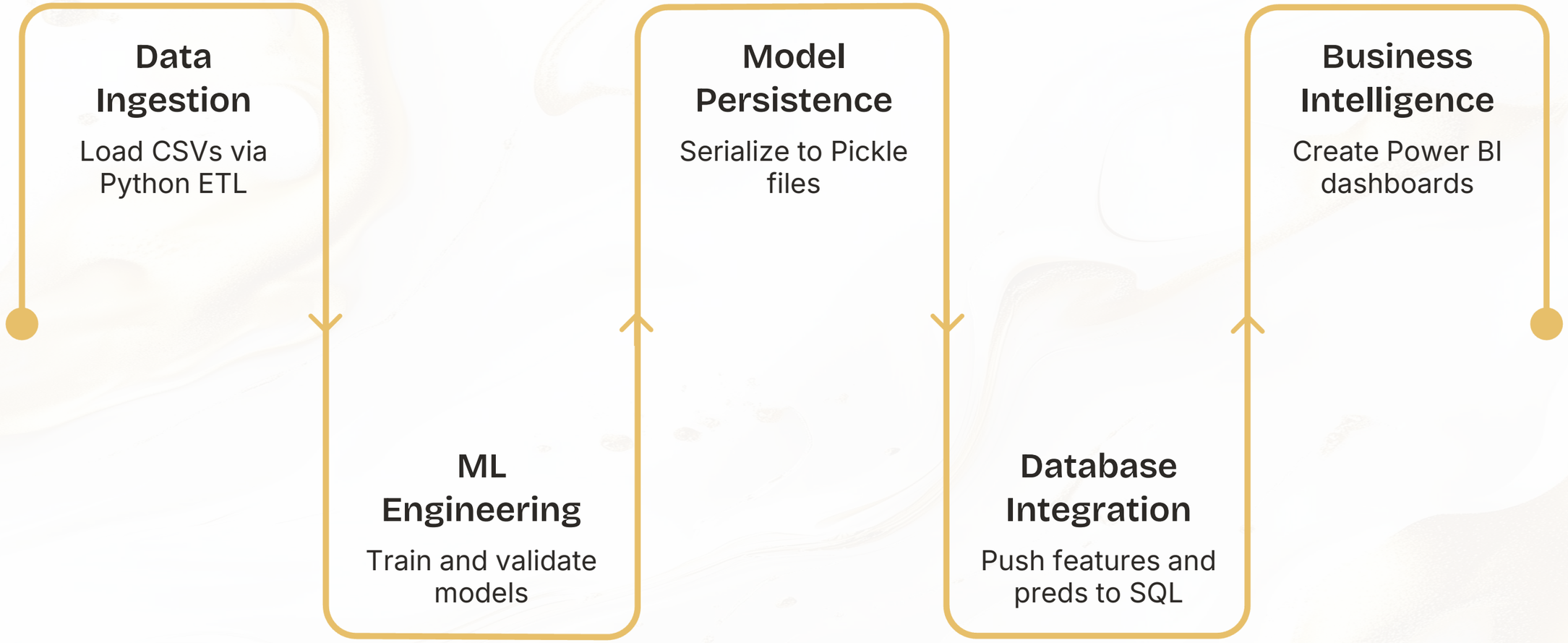
The Business Challenge

From Reactive to Proactive

In SaaS, customer retention is the primary driver of profitability. This project transitions from reactive churn tracking to **proactive churn prediction** — identifying at-risk accounts before they cancel, empowering targeted retention strategies.

Core Technologies: Python (Pandas, Scikit-Learn), Pickle, SQL, Power BI, Jupyter Notebooks

End-to-End Pipeline Architecture



Each phase builds on the last — from raw CSV ingestion through Python, to a deployed Pickle model, SQL analytics layer, and executive Power BI dashboards.

Phase 1 & 2: Data Ingestion, Auditing & Wrangling

Data Audit

Profiled dimensions, variable types, and distributions. Detected anomalies, missing values, and outliers in raw enterprise data.

Imputation & Cleaning

Median imputation for skewed numerics, mode replacement for categoricals — preserving data integrity throughout.

Feature Engineering

Derived ARPU, Support Ticket Frequency, and Engagement Score — capturing nuanced behaviors of churning customers.

Encoding & Scaling

One-Hot Encoding for categorical text; numerical scaling to prevent algorithmic bias in downstream models.

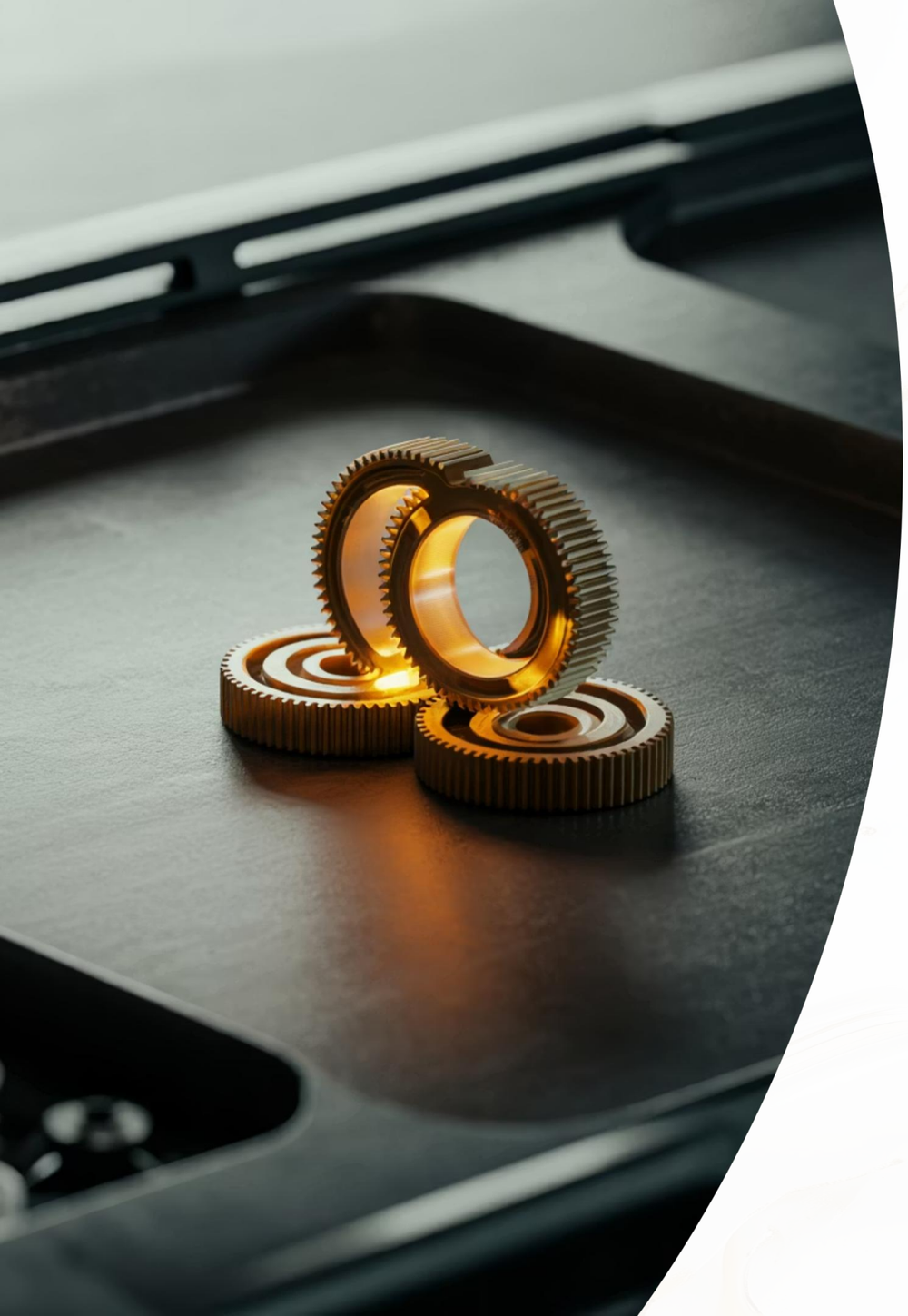
Phase 3: Machine Learning & Predictive Modeling

Model Selection & Optimization

Trained and evaluated multiple classifiers, then fine-tuned hyperparameters to balance **Precision** (minimizing false alarms) and **Recall** (catching every at-risk customer).

Feature importance analysis revealed that **support response times** heavily influence cancellation decisions.

The business now has mathematical churn probabilities — not guesswork — backing every retention campaign.



Phase 4: Scalable Model Deployment via Pickle

Model Serialization

The fully trained model is "dumped" into a lightweight `saasmodel.pkl` file — no retraining required per prediction.

Operational Efficiency

Internal systems and APIs load the model into memory in **milliseconds**, saving extensive computational resources.

Real-Time Readiness

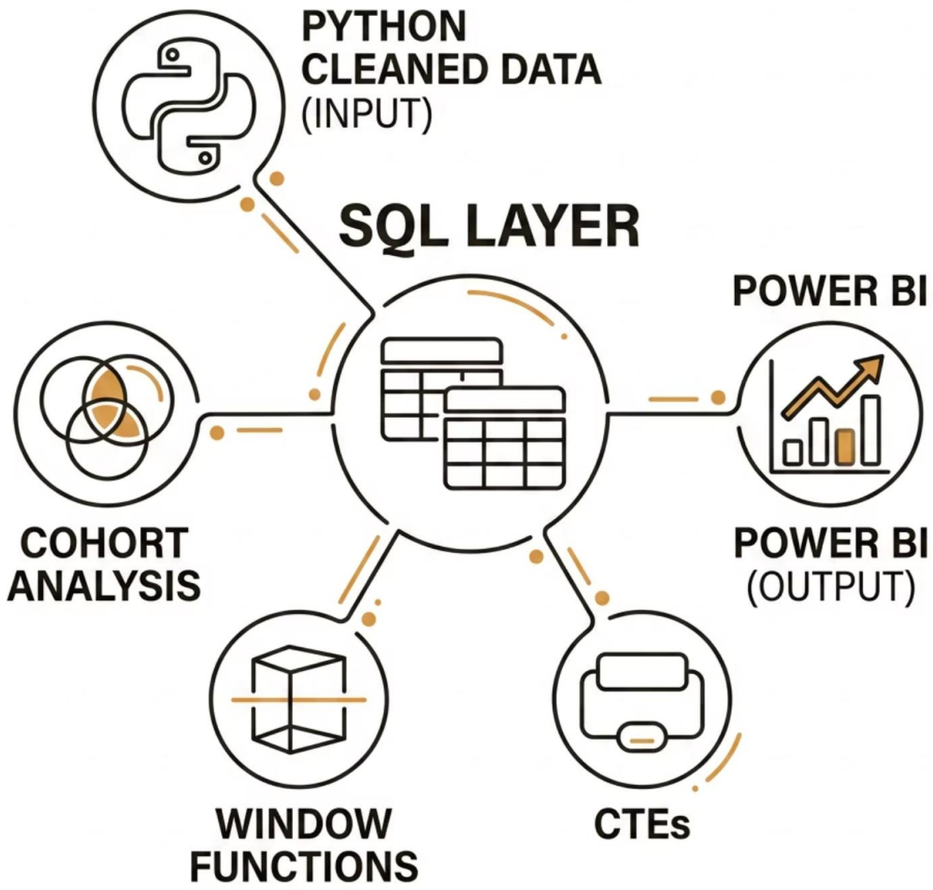
Ingests live daily user data and outputs immediate churn probability scores, feeding automated marketing and Customer Success alerts.

Phase 5: SQL Database Integration

Query Engineering

Advanced SQL scripts using **JOINs, CTEs, and Window Functions** aggregate customer behavior at scale.

- Cohort analysis by acquisition month, subscription tier, and engagement level
- Indexed data warehouse structure for high-performance BI queries
- Single source of truth democratizing data for non-technical stakeholders



Phase 6: Power BI Executive Dashboards

KPI Tracking

Global Churn Rate, MRR at risk, and Customer Lifetime Value displayed as top-level executive metrics.

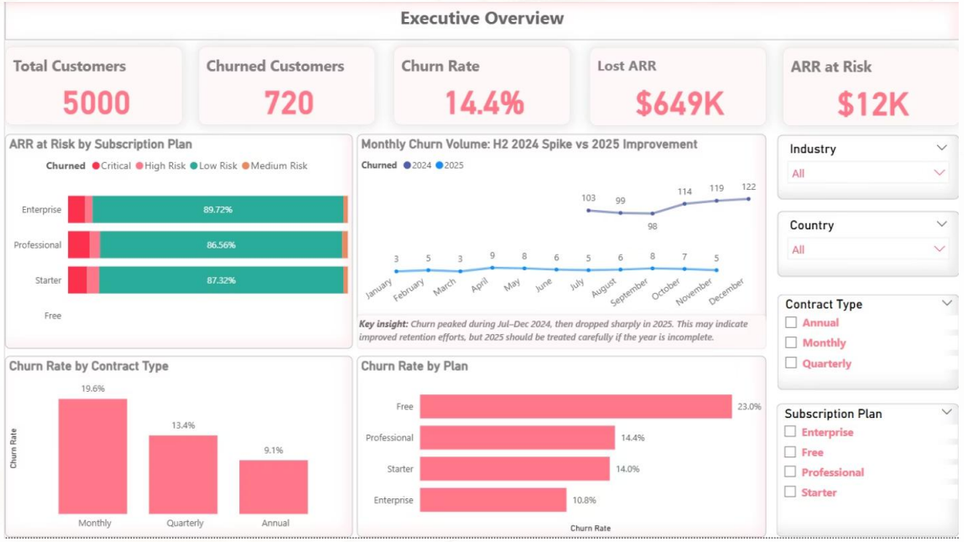
Interactive Slicers

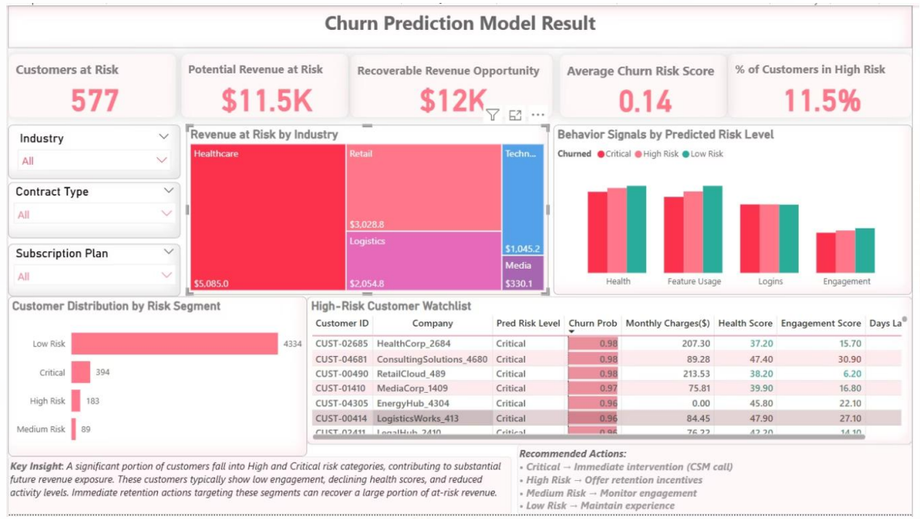
Dynamic filters to drill into churn by geography, company size, and product feature usage.

Predictive Visualization

ML model outputs displayed as live "High-Risk Customer Lists" — updating dynamically per user selection.

Strategic Business Recommendations





1

Automated CSM Outreach

Accounts scoring **>75% churn probability** via the Pickle model should automatically trigger a Customer Success Manager workflow.

2

Onboarding Optimization

SQL cohort analysis revealed high drop-offs in **Month 2**. Revamping the 30-day onboarding sequence is mathematically proven to improve retention.

3

Feature Adoption Focus

Users engaging with core "sticky" features churn at significantly lower rates. Drive adoption via **in-app prompts** targeting these specific tools.

Business Value Delivered

Shifted Paradigm

Proactive vs. Reactive — predicting churn before it happens.

Production Ready

Machine Learning model serialized and deployable into existing infrastructure immediately.

Executive-Facing

Power BI dashboards translate complex data science into click-and-explore C-suite intelligence.

> This infrastructure does not merely explain the past — it actively predicts the future, delivering immediate financial impact to any data-driven SaaS organization.